

WORLD-MODEL TRANSFER FOR DATA-EFFICIENT TIME-SERIES FORECASTING

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Paper under double-blind review

ABSTRACT

Zero-shot time-series forecasters are normally bought with data: the strongest open models pretrain on corpora of tens to hundreds of billions of observations. We show that a video-pretrained backbone plus a transfer framework designed for forecasting reaches the same accuracy band for a small fraction of that cost. The obstacle is that the transfer does not happen on its own — continuing a video model’s masked-pixel objective on rendered series lowers that objective while raising forecast error — so we rebuild the adaptation stage around the forecasting task. A series is rendered one period per frame, which makes motion across frames the evolution from one period to the next; the model predicts a masked block of future frames from the visible past alone; a differentiable read-out decodes those frames back into values, and the loss is taken there rather than on pixels; and training samples exactly the frame scales the inference rule will use. After 60k optimizer updates on 654M observations — 18.9 hours on one A40, about 0.3% of the archive and 1/27 of the sample presentations behind the strongest visual baseline — a single checkpoint under a single rule attains a mean MSE of 0.285 over six standard long-sequence datasets and four horizons, the lowest of the models we compare, against 0.289 for VisionTS++ large; its mean MAE, 0.328, is behind that model’s 0.326. Holding everything but the initialisation fixed, the video prior is worth 3–5% at every horizon at a 5k-update budget, and an ImageNet prior catches up by 20k: what video pretraining buys in this regime is adaptation efficiency, which is precisely the resource the low-budget setting is short of.

1 INTRODUCTION

Zero-shot forecasters are trained once and applied to datasets they have never seen. The open models that do this well — Moirai (Woo et al., 2024), Chronos (Ansari et al., 2024), Time-MoE (Shi et al., 2025), TimesFM (Das et al., 2024) — reach that ability by pretraining on time-series corpora of tens to hundreds of billions of observations. The cost is the point: it decides who can build such a model and how often it can be rebuilt when the data distribution moves. We ask what the accuracy ceiling looks like at a small fraction of that budget, and we get there by starting from a backbone that already knows something about how signals evolve.

The visual route shows the idea is viable. VisionTS (Chen et al., 2025) renders a series as an image and lets a frozen ImageNet masked autoencoder (He et al., 2022) inpaint the missing portion; VisionTS++ (Shen et al., 2025b) continues that backbone’s pretraining on rendered series with a multi-quantile head and reports the strongest published zero-shot numbers on the standard long-sequence protocol. Both inherit a prior over spatial structure from a model that has never seen anything change over time. Video pretraining offers a different inheritance — representations built from sequences, and in the predictive variants an explicit forward model (Tong et al., 2022; Assran et al., 2025) — and the useful question about it is not whether it raises the ceiling but whether it lowers the cost of reaching one: *does a video-pretrained backbone reach a given forecasting accuracy under a smaller time-series adaptation budget?*

Getting an answer takes more than swapping the backbone. A numeric series has to become a video whose axes carry temporal meaning; masked reconstruction, the objective the backbone arrives with, recovers a token from neighbours on both sides, which is interpolation rather than forecasting; and the backbone minimises a gray level while the task scores a number, with an untrained de-

coder between them. The consequence is measurable: continuing VideoMAE’s native objective on a 654M-observation corpus of rendered series lowers its held-out pixel loss monotonically and raises forecast error monotonically across the checkpoints we evaluated, 0.339 at 5k updates to 0.381 at 176k (Section 4.6). More of that pretraining makes the forecaster worse.

We therefore keep the video weights and rebuild the adaptation stage around forecasting (Figure 1). **Period-aligned rendering** puts one period on the horizontal axis of a frame and value on the vertical axis, so that the frame axis is time measured in periods and a video backbone’s extrapolation axis lines up with the series’ own. **Future-block prediction with a differentiable read-out** masks a trailing block of frames, predicts it from the visible past only, decodes the predicted frames back into values, and takes the loss on those values, which puts the pixel-to-number decoder inside the optimisation instead of after it. **Temporal-scale alignment** trains on exactly the periods-per-frame scales the inference rule selects, so the model is never asked at test time for a resolution it was not trained at. These are design choices for a transfer problem, not three independent inventions: Section 2 places each against its nearest existing method, and Section 4.5 reports the ablation that each is supported by.

The result is a position on the accuracy–resource trade-off rather than a large accuracy jump. Sixty thousand optimizer updates at an effective batch of 32 over 654M observations — 18.9 hours on one A40, about 0.3% of the LOTSA archive and 1/27 of the sample presentations behind VisionTS++ — give one checkpoint that, under one inference rule, reaches a mean MSE of 0.285 over six standard datasets and four horizons: the lowest of the models we compare (VisionTS++ large 0.289, base 0.291, VisionTS 0.309), by a small margin. On mean MAE we are behind, 0.328 against 0.326, and per cell against VisionTS++ base we take 17 of 24 on MSE but 11 on MAE. The efficiency statement is the substantive one; the accuracy statement is that we stay in the leading band while making it.

Two controlled experiments locate the video prior’s contribution. Changing only the starting weights, it leads an ImageNet initialisation by 3–5% at every horizon at 5k updates, and by 20k the image initialisation has caught up; separately, removing the pretrained weights altogether costs 14% under an otherwise identical trainer. What video pretraining supplies here is adaptation efficiency — the binding constraint in this regime — and not a higher ceiling. We ran that comparison at 5k and 20k updates and not at the 60k of the reported system, so the efficiency claim is the one the study supports; we do not claim the headline number depends on the video initialisation.

Contributions.

- A transfer framework that makes a video-pretrained backbone usable as a zero-shot forecaster: period-aligned video rendering, future-block prediction supervised through a differentiable numeric read-out, and a training scale distribution matched to the inference rule (Section 3).
- Evidence that this reaches the accuracy band of the strongest published zero-shot models at 0.3% of their time-series corpus, 1/27 of their sample presentations and 18.9 GPU-hours, with the budget reported in separated units and no compute claim (Sections 4.2–4.3).
- A measurement of what the video prior is worth and where it stops: a matched-size, matched-recipe initialisation study with a random-init control at two adaptation budgets (Section 4.4).
- A diagnosis of why the naive route fails — the backbone’s own objective and forecasting accuracy move in opposite directions as pretraining continues — which is the motivation for the framework and a caution for anyone continuing a vision objective on rendered series (Section 4.6).

2 RELATED WORK

Time-series foundation models. Moirai (Woo et al., 2024), Chronos (Ansari et al., 2024), Time-MoE (Shi et al., 2025), Timer (Liu et al., 2024), TimesFM (Das et al., 2024) and MOMENT (Goswami et al., 2024) pretrain on large time-series corpora with tokenisation, quantisation or patching designed for the modality, and are evaluated zero-shot on held-out datasets. They set both the accuracy we compare against and the resource bar we are trying to undercut.

Forecasting with visual backbones. VisionTS (Chen et al., 2025) folds a series into a periodicity-aligned image and inpaints the right-hand portion with a frozen ImageNet masked autoencoder (He et al., 2022). VisionTS++ (Shen et al., 2025b) keeps that construction, continues pretraining on LOTSA, and adds colourised multivariate rendering and nine quantile heads trained with a pinball objective. Two observations from this line matter for us. The visual prior that survives the transfer appears to be small — SVTime (Shen et al., 2025a) matches VisionTS with a 215k-parameter model and OccamVTS (Lyu et al., 2026) distils a vision forecaster into 1% of its parameters. And VisionTS++ already supervises numerical forecasts directly: our objective-mismatch result (Section 4.6) is a statement about VideoMAE’s native masked-pixel objective continued on rendered series, the configuration we tested, not a criticism of the image-based systems we compare against.

Video pretraining and future prediction. VideoMAE (Tong et al., 2022) reconstructs masked spatiotemporal tubes, so its temporal structure is implicit rather than a forward model; V-JEPA (Bardes et al., 2024) and V-JEPA 2 (Assran et al., 2025) predict latent representations of masked regions and are used for planning, closer to the world-model formulation of LeCun (2022) and Hafner et al. (2025). We use both as initialisations. The sense in which this paper’s training stage is a world model is the operational one stated in the abstract: it maps an observed past to a predicted future state of the series and is scored on that prediction. That is the whole of the definition we use, and the title should be read with it — we test no claim about action conditioning, planning, learned simulation or physical understanding, and we do not describe the VideoMAE checkpoint itself as any of those things.

What is new here, component by component. Each element of Section 3 is a change relative to a specific existing practice. *Rendering:* VisionTS folds a series into one image whose rows are periods; we render one period per *frame*, so the period index becomes the model’s temporal axis and the masked region is a block of future frames rather than a region of a still image. It is the representation all our arms run in; no experiment isolates it from frame-based prediction. *Supervision:* value-space losses are standard in forecasting and VisionTS++ trains a quantile objective over values. What is new is putting a differentiable pixel-to-value read-out *inside* a masked-video pretraining objective, which converts a video backbone’s reconstruction loss into a forecasting loss without touching the backbone; Table 4 measures it against pixel-only supervision under matched conditions. *Scale alignment:* multi-scale training is common, but here the training distribution over periods-per-frame is set equal to the one the inference rule draws from — a consistency property, part of the recipe rather than an independently ablated component.

Transfer regularisation. L2-SP (Li et al., 2018) decays weights toward their initial values rather than toward zero. It is an existing method used here as an ablation arm: it helps at 16-frame clips and is inert in the configuration of our main model (Appendix C).

3 A TRANSFER FRAMEWORK FROM VIDEO PRETRAINING TO FORECASTING

Overview. The framework is one pipeline, run in two directions (Figure 1):

observed series → context-only normalisation → period-aligned frames → visible past / masked future →
video backbone → predicted frames → numeric read-out → {forecasting loss, forecast}.

During time-series continual pretraining the last step produces a loss; at test time it produces the forecast. The design question is what has to be true of the middle of that pipeline for weights learned on natural video to be worth anything at the end of it, and the three subsections below give our three answers: a representation in which frames carry temporal evolution (§3.2), a training signal that is a forecast rather than a reconstruction (§3.3), and a match between the temporal scales seen in training and those used at inference (§3.4).

3.1 THREE STAGES, AND WHAT ZERO-SHOT REFERS TO

(i) Video pretraining is inherited: a public VideoMAE-B checkpoint trained on Kinetics-400 (Tong et al., 2022), and in the initialisation study V-JEPA checkpoints (Assran et al., 2025). **(ii) Time-series continual pretraining** is the stage this paper designs, on a corpus containing none of the

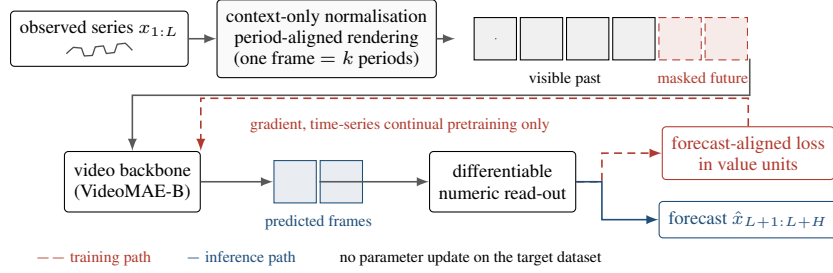


Figure 1: The transfer framework. A series is normalised with the statistics of its visible context only and rendered so that one frame holds k periods, which puts value on the vertical axis and phase on the horizontal one; the trailing frames are masked. A video backbone predicts those frames, and a differentiable read-out decodes them back into values. During time-series continual pretraining the loss is taken on the decoded values (Section 3.3); at test time the same read-out emits the forecast and no parameter is updated on the target dataset. The backbone enters this pipeline with weights from video pretraining, which is the only stage that sees video.

evaluation datasets (§3.6). (iii) **Zero-shot evaluation** applies the resulting checkpoint with no parameter update and no per-dataset tuning. Design decisions and checkpoint selection used held-out pretraining data only.

3.2 A. PERIOD-ALIGNED VIDEO REPRESENTATION

A series is turned into a video, not into an image. Let P be the dominant period, taken from the sampling frequency as in Chen et al. (2025), and let k be the number of periods packed into one frame. A clip of F frames covers FkP observations: frame f holds the kP consecutive observations of periods $[fk, (f+1)k)$, drawn side by side as an area chart in a 224×224 frame. Within a frame the horizontal axis is position inside that block of k periods — phase, when $k = 1$ — and the vertical axis is value; across frames the axis is time measured in blocks of k periods. The kP samples are spread over the 224 columns by a fixed integer partition, so each sample owns $\lfloor 224/kP \rfloor$ or $\lceil 224/kP \rceil$ adjacent columns and the mapping is invertible by construction. Consecutive frames therefore differ by whatever changed from one period to the next — level, amplitude, shape — and a seasonal series renders as an almost static video while a drifting or regime-switching one renders as a moving one.

This is what separates the representation from image folding. Folding a series into a 2-D image also aligns periods, but the model then sees one static picture to complete. Here the period axis is the model’s *time* axis — the axis a video backbone was pretrained to extrapolate along — and the masked region is a block of future frames.

Concretely, values are standardised with the mean and standard deviation of the visible context only, clipped to $z \in [-3, 3]$, and mapped to a column height

$$h = 0.5 + 0.4 \frac{z}{3} \in [0.1, 0.9], \quad (1)$$

with the column filled below the boundary row $\text{round}((1-h)(\text{IMG}-1))$ and background above it. The drawing constants and the grid are given in Appendix A. The main model uses $F = 32$ frames with a tubelet size of 2, giving 16 temporal positions; F is a context-length setting rather than a contribution, and we report what changes with it in Appendix C.

3.3 B. FORECAST-ALIGNED PREDICTION AND SUPERVISION

Predict a future block, not a random subset. The trailing h_p frames of the clip are masked and must be predicted from the visible past. This is the formulation the title refers to: the model maps an observed past to a predicted future state of the series and is scored on that prediction, in the sense of LeCun (2022), and unlike the backbone’s native masked-reconstruction objective it never sees a neighbour on the far side of what it is predicting. With probability 0.3 a uniformly drawn

1... $(F - h_p - 8)/2$ leading tubelets are additionally hidden, so that shorter contexts are trained on as well; those hidden tokens are predicted but excluded from the loss. Thirty per cent of batches instead retain the backbone’s original tube masking at a 0.75 ratio, as a regulariser. The two batch types are supervised differently: forecast batches take both loss terms on the future tokens only, while tube batches take the pixel term on all masked tokens and carry no value term (Appendix A).

Supervise in value space. A predicted frame is a picture, and the quantity the task is scored on is a number, so the decoder that converts one into the other belongs inside the loss. For a predicted frame $\hat{x}$ we compute a differentiable soft occupancy per pixel and sum it down each column,

$$s(\hat{x}_{r,c}) = \text{clamp}\left(\frac{g_{\text{grid}} - \hat{x}_{r,c}}{g_{\text{grid}} - g_{\text{fill}}}, 0, 1\right), \quad \hat{h}_c = 1 - \frac{\text{IMG} - \sum_r s(\hat{x}_{r,c})}{\text{IMG} - 1}, \quad (2)$$

which inverts the drawing rule of Equation 1 up to rounding: the thresholds are chosen so that background and grid pixels contribute zero and filled pixels contribute one, and partially filled boundary pixels contribute fractionally. Inverting Equation 1 gives $z = (h - 0.5) \cdot 3/0.4$, and the training objective is

$$\mathcal{L} = \text{MSE}(\hat{x}, x) + \lambda \left(\frac{3}{0.4}\right)^2 \text{MSE}(\hat{h}_c, h_c), \quad \lambda = 0.1. \quad (3)$$

The constant converts height units into the units of the context-standardised series, so the second term is a squared error on values rather than on gray levels. It is computed column-wise and is therefore independent of P and of how many periods a frame holds. The pixel term is retained because early in training the read-out of a blurry prediction carries little signal.

One clarification about units: this error is in the *context-instance* z-scale, since each window is standardised by its own visible statistics, while the benchmark reports errors in a scale fitted on the target dataset’s training split. The two differ by the ratio of those standard deviations, which varies per window. The objective is therefore an error on values rather than on pixels, which is the property we need, but it is not numerically identical to the benchmark metric.

3.4 C. TEMPORAL-SCALE ALIGNMENT

A forecaster must cover horizons from one period to many, and in this representation the horizon a single predicted block can cover is $h_p k P$. Inference therefore varies k with the requested horizon (§3.5), and training must cover the same range or the model meets resolutions it never saw. We sample $k \in \{1, 2, 4\}$ during training with probabilities 0.5/0.3/0.2, which is the distribution the inference rule draws from. This is a training–inference consistency property of the framework rather than a generic augmentation: the set of scales is determined by the inference rule, not chosen for diversity.

3.5 INFERENCE

Given the period P and the requested horizon H , a closed-form rule selects the number of periods per frame and the number of forward passes. The rule contains no dataset identity and is applied unchanged to all 24 cells, but its two thresholds were chosen by comparing configurations on development runs over four of the six datasets, so it is dataset-agnostic in form rather than selected without reference to the benchmark; we flag this as a limitation in Section 5. Short horizons are covered by a single predicted block whose outputs are averaged over the admissible scales; long horizons increase k and, when one block still cannot span H , roll the prediction forward autoregressively. The resulting forecast is truncated to H . Multi-scale averaging and rollout both cost additional forward passes, so the system is *single-checkpoint* and *single-rule* but not *single-pass*; Appendix D gives the configuration and the pass count for each of the 24 evaluation cells.

3.6 PRETRAINING CORPUS

The corpus is built from LOTSA (Woo et al., 2024): 98 subset directories, of which 74 contribute at least one qualifying series, giving 184,013 series and 654.2M observations; a 5% holdout leaves 621.9M reachable by training. Subsets are drawn in proportion to the square root of their size, and with probability 0.2 a window is replaced by a synthetic series generated on the fly, so the synthetic data sits on top of the 654.2M rather than inside it. Sampling is with replacement, so

Table 1: Zero-shot long-sequence forecasting. Each entry averages the four horizons $H \in \{96, 192, 336, 720\}$; MSE and MAE are computed on standardised values. Our model uses one checkpoint and one inference rule for all 24 cells. Baselines are quoted from Shen et al. (2025b) and Chen et al. (2025); ‘–’ means the source does not report that dataset. **Bold**: best. Underlined: second best. The last column counts first places over the twelve dataset columns.

Model	ETTm1		ETTm2		ETTh1		ETTh2		ECL		Weather		Avg		1 st
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
Ours	0.352	0.369	0.241	<u>0.302</u>	0.404	<u>0.416</u>	0.335	0.376	0.163	0.254	0.215	0.252	0.285	0.328	6
VisionTS++ (ViT-L)	<u>0.354</u>	0.369	<u>0.244</u>	0.298	0.403	0.418	0.327	0.365	<u>0.181</u>	<u>0.264</u>	<u>0.226</u>	<u>0.243</u>	<u>0.289</u>	0.326	4
VisionTS++ (ViT-B)	<u>0.360</u>	<u>0.372</u>	<u>0.244</u>	0.298	0.402	<u>0.416</u>	<u>0.333</u>	<u>0.370</u>	<u>0.184</u>	<u>0.265</u>	<u>0.222</u>	0.241	<u>0.291</u>	<u>0.327</u>	2
VisionTS	0.374	<u>0.372</u>	0.282	0.321	0.390	0.414	<u>0.333</u>	0.375	0.207	0.294	0.269	0.292	0.309	0.345	2
Moirai _S	0.448	0.410	0.300	0.341	0.400	0.424	0.341	0.379	0.233	0.320	0.242	0.267	0.327	0.357	–
Moirai _B	0.382	0.388	0.272	0.321	0.434	0.439	0.346	0.382	0.188	0.274	0.238	0.261	0.310	0.344	–
Moirai _L	0.390	0.389	0.276	0.320	0.510	0.469	0.354	0.377	0.188	0.273	0.260	0.275	0.330	0.350	–
Chronos _S	0.640	0.500	0.349	0.380	0.545	0.472	0.424	0.430	0.220	0.284	0.300	0.318	0.413	0.397	–
Chronos _B	0.646	0.500	0.310	0.350	0.591	0.468	0.406	0.411	0.215	0.279	0.293	0.315	0.410	0.387	–
Chronos _L	0.556	0.465	0.295	0.338	0.589	0.466	0.455	0.427	0.204	0.274	0.279	0.306	0.396	0.379	–
Time-MoE _S	0.394	0.416	0.318	0.366	0.400	0.424	0.367	0.404	–	–	0.266	0.297	–	–	–
Time-MoE _B	0.376	0.406	0.316	0.361	<u>0.394</u>	0.420	0.405	0.415	–	–	0.270	0.300	–	–	–
Timer	0.487	0.457	0.316	0.371	0.444	0.457	0.358	0.407	–	–	0.304	0.331	–	–	–
TimesFM	0.433	0.419	0.328	0.347	0.473	0.444	0.392	0.406	–	–	–	–	–	–	–
MOMENT	0.670	0.537	0.317	0.366	0.684	0.566	0.362	0.410	0.765	0.687	0.294	0.326	0.515	0.482	–

the 60k-update run draws 1.92M windows of ≈ 2.9 k observations each, about seven passes over the reachable corpus, while the 5k-update run of Section 4.4 sees less than one.

Two facts about the corpus deserve to be stated rather than buried. No ETT data of any kind is in it. But it is not domain-filtered: it contains Monash `weather` (a daily Australian dataset, not the 10-minute Jena set used as the LSF weather benchmark) and Monash `traffic_hourly`, which is the same freeway-occupancy source as the LSF traffic benchmark. Traffic is therefore excluded from every table in this paper. An exact-match leakage screen over the benchmark files returned no matches, but it was run against an earlier, domain-excluded build of the corpus rather than this one, so we report it as a screen we have not yet repeated on the corpus actually used (Appendix A).

4 EXPERIMENTS

The experiments answer four questions in order: the accuracy the framework reaches under a small time-series pretraining budget (§4.2–§4.3); what the video initialisation contributes and where that stops (§4.4); what the transfer design contributes (§4.5); and why the naive alternative fails (§4.6).

4.1 SETUP

Protocol. We follow the zero-shot long-sequence protocol of Chen et al. (2025): Time-Series-Library splits (Wu et al., 2023) (ETT 12/4/4 months, others 70/10/20), a `StandardScaler` fitted on the target dataset’s training portion, all channels, stride-1 test origins, $H \in \{96, 192, 336, 720\}$, MSE/MAE on standardised values, and their per-dataset context lengths and periods (Appendix A). Running their released checkpoint through our harness reproduces their published numbers to three decimals (ETTh1 at $H=96$: 0.3527/0.3833 against 0.353/0.383). One checkpoint and one inference rule serve all 24 cells, with no parameter updated on any evaluation dataset.

Ablations and baselines. Ablations and initialisation studies use five datasets (ETT $\times 4$ and weather) at three horizons; the six-dataset four-horizon grid is reserved for the main table. Five- and six-dataset means are never mixed, and every table says which it reports. Baselines are quoted from Shen et al. (2025b) and Chen et al. (2025); we did not retrain them.

4.2 LOWEST MEAN MSE OF THE MODELS COMPARED, BEHIND ON MEAN MAE

Table 1 reports four-horizon averages. Our model has the lowest mean MSE of the models compared, 0.285 against 0.289 for VisionTS++ large and 0.291 for its base model, and takes the most first

Table 2: Time-series pretraining budget in the units each paper reports. Sample presentations are optimizer updates $\times$ effective batch, i.e. how many rendered windows the model was shown; they are not a compute measure, since the two systems differ in model size and in the number of time points and visual tokens per sample. The upstream video or image pretraining that both inherit is excluded, as is any FLOP or wall-clock comparison across systems. VisionTS++ figures are as reported in Shen et al. (2025b). Takeaway: the reported accuracy is obtained from $\approx 0.3\%$ of the archive and $\approx 1/27$ of the sample presentations, in 18.9 hours on one GPU.

	Ours (main)	Ours (20k)	VisionTS++
Time-series corpus (observations)	654.2M	654.2M	LOTSa archive, 231B
reachable after the 5% holdout	621.9M	621.9M	not reported
Optimizer updates	60,000	20,000	100,000
Effective batch (windows/update)	32	32	512
Sample presentations	1.92M	0.64M	51.2M
Backbone	ViT-B (87M)	ViT-B (87M)	ViT-B / ViT-L
Effective passes over the corpus	≈ 6.8	≈ 2.3	not reported
Measured training time	18.9 h, 1 $\times$ A40	6.3 h, 1 $\times$ A40	not reported

places across the twelve dataset columns. The margin comes from electricity (0.163 against 0.181 for VisionTS++ large, a 10% reduction) and weather (0.215 against 0.222 for VisionTS++ base and 0.226 for large); ETTh1 goes to VisionTS (0.390 against 0.404) and ETTh2 to VisionTS++ large (0.327 against 0.335), both hourly sets at long horizons, where our 32-frame window has to trade phase resolution for coverage.

The metric picture is mixed and we state it plainly: mean MAE is 0.328 against VisionTS++ large’s 0.326, so on that metric we are behind, and per cell against the same-size VisionTS++ base we take 17 of 24 on MSE but only 11 on MAE (Appendix B). The claim this table supports is that the framework reaches the leading band on MSE, not that it dominates it.

At 20k updates instead of 60k the same recipe reaches 0.288/0.331, still first on mean MSE here at a third of the training.

Averaged over the six datasets, error grows from 0.223/0.281 at $H=96$ to 0.349/0.377 at $H=720$; Appendix B gives every cell against both baselines that publish per-horizon numbers.

4.3 18.9 GPU-HOURS ON 0.3% OF THE ARCHIVE

Table 2 states the resources on both sides in the units each paper reports. The main model is 60k optimizer updates at an effective batch of 32 over a 654.2M-observation corpus, and it took 18.9 hours on a single A40. Against VisionTS++’s reported 100k steps at batch 512 over the 231B-observation LOTSA archive, that is $\approx 1/27$ of the sample presentations and $\approx 0.3\%$ of the corpus.

We keep these quantities separate rather than collapsing them into one efficiency number, because they are not interchangeable: the models differ in size, a sample contains a different number of time points and visual tokens under the two renderings, and the baseline’s hardware and wall-clock are not reported. For the same reason we make no FLOP or speed-up claim. The accounting also excludes the upstream video pretraining we inherit, exactly as the image-based systems inherit ImageNet pretraining; what it measures is the cost of the time-series stage, which is the stage a practitioner would have to pay for again.

4.4 WHAT THE VIDEO INITIALISATION IS WORTH, AND AT WHICH BUDGET

Table 3 answers this with two controls, split into panels because only within a panel is the initialisation the sole difference. In the upper panel — same rendering, objective, corpus, schedule and inference rule, ViT-B either way — the video initialisation leads the ImageNet one by 3–5% at every horizon after 5k updates. The lower panel adds the random-init control, which is the cleanest statement about pretraining as such: the same ViT-L architecture under the same trainer, data and schedule reaches 0.280 at $H=96$ with no pretrained weights against 0.245 with them, a 14% gap from the weights alone. That randomly initialised ViT-L is also worse than the pretrained ViT-B a

Table 3: Initialisation study: five-dataset mean MSE (ETT $\times$ 4 and weather) at $H=96/336/720$, averaged over datasets. *Within* a panel the arms differ only in the weights the model starts from, with identical rendering, objective, corpus, schedule and inference rule; *across* panels they differ in trainer and backbone size as well, so the panels are not comparable to one another. Two seeds are listed where both were run. Takeaway: the video prior is worth 3–5% at a 5k-update budget and is matched by the image prior by 20k, while random initialisation of the same architecture is worse than an image-pretrained backbone $3.5\times$ smaller.

Initialisation	Pretraining signal	Params	5k updates	20k updates
<i>Masked-pixel trainer, ViT-B: identical apart from the initialisation</i>				
ImageNet MAE-B	images	87M	0.264 / 0.350 / 0.395	0.244 / 0.329 / 0.373
				0.245 / 0.330 / 0.373
VideoMAE-B	video, reconstructive	87M	0.250 / 0.337 / 0.384	0.245 / 0.330 / 0.379
<i>Latent-prediction trainer: different objective and architecture, not an initialisation control</i>				
V-JEPA 2.1-B	video, predictive	87M + 23M	0.252 / 0.335 / 0.382	not run
V-JEPA 2-L	video, predictive	300M + 22M	0.245 / 0.329 / 0.379	not run
V-JEPA 2-L, random	none	300M + 22M	0.280 / 0.369 / 0.418	not run

Table 4: Ablation of the transfer design: five-dataset mean MSE (ETT $\times$ 4 and weather) at three horizons, averaged over datasets, with 32-frame clips and an identical corpus, masking schedule, optimiser schedule and inference rule; only the listed component changes. Takeaway: replacing the value-space read-out with pixel-only supervision costs 5–8%, with the largest cost at the longest horizon, while prior anchoring changes nothing at this clip length.

Configuration	$H=96$	$H=336$	$H=720$
Full framework, 20k updates	0.245	0.330	0.379
pixel-only supervision (no value-space term)	0.257	0.347	0.411
no prior anchoring	0.244	0.330	0.378
Full framework, 5k updates	0.250	0.337	0.384
Full framework, 60k updates	0.242	0.329	0.377

third its size in the upper panel, which is suggestive about capacity, but that particular comparison crosses the two trainers and we do not treat it as controlled.

Two boundaries belong with this result, and the first is consequential. The advantage is budget-dependent: at 20k updates the image initialisation has caught up and is slightly ahead at the longest horizon, so what video pretraining supplies is adaptation efficiency rather than a higher ceiling. The reported 60k system sits past that crossover and we did not train a 60k image-initialised counterpart, so this study does not establish that the headline number requires the video initialisation — it establishes what the video prior is worth at the budgets we ran the comparison at. And the panels are not comparable to each other — the lower arms use a different trainer (latent prediction with an added predictor, an EMA target encoder, no tube-masked batches) on a larger backbone — so nothing here shows a predictive video objective beating a reconstructive one; at ViT-B the two are within 0.002. The image-initialised arm is an initialisation control, not a reimplement of VisionTS++.

4.5 VALUE-SPACE SUPERVISION ACCOUNTS FOR THE MEASURED GAIN OVER PIXEL-ONLY TRAINING

Table 4 varies the supervision and the budget with everything else fixed. Replacing the value-space term with pixel-only supervision — same masking, rendering, corpus, schedule and inference rule — costs 5% at $H=96$, 5% at $H=336$ and 8% at $H=720$. Under this matched comparison the read-out term accounts for the measured gain of the recipe, and its value grows with the horizon, which is where more of the output is generated rather than carried over from the visible context. Prior anchoring (Li et al., 2018) is inert at the main model’s 32-frame clip length and matters only at 16 frames (Appendix C); we report it as an inherited regulariser that this configuration does not need.

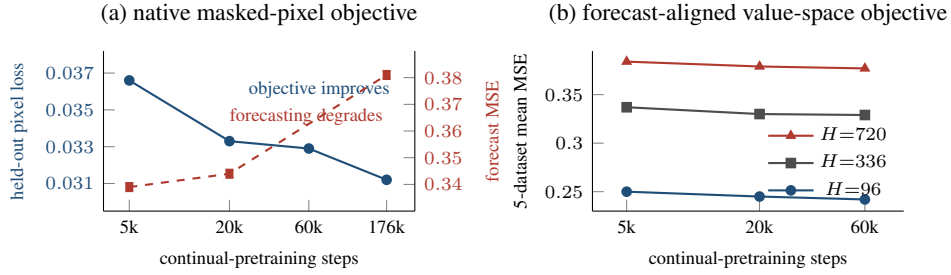


Figure 2: The same backbone and corpus under two objectives. (a) VideoMAE’s native masked-pixel objective: its own held-out loss falls monotonically while forecast error rises monotonically over a $35\times$ range of budget. The 60k checkpoint was not evaluated downstream. (b) The forecast-aligned value-space objective of Section 3.3: the curve is monotonically improving at every horizon. Panel (a) uses 16-frame clips and panel (b) 32-frame clips; the ablation in Table 4 compares the two objectives at matched clip length.

Under this objective the budget curve is monotonically improving: $0.250 \rightarrow 0.245 \rightarrow 0.242$ at $H=96$ for 5k, 20k and 60k updates. Scaling the other two knobs did not help — a larger effective batch is equal or worse at every horizon and a $4\times$ larger in-domain corpus ties ours — Appendix G.

4.6 THE NATIVE OBJECTIVE AND FORECASTING ACCURACY MOVE IN OPPOSITE DIRECTIONS

Figure 2a is the observation that motivated the design. Continuing VideoMAE’s native masked-pixel objective on the rendered corpus lowers its held-out pixel loss monotonically over the four checkpoints we evaluated ($0.0366 \rightarrow 0.0312$) while forecast error rises monotonically over the three evaluated downstream ($0.339 \rightarrow 0.381$). Per cell the 176k model is 5–60% worse than the 20k one, worst at long horizons (ETTh2 at $H=720$: $0.429 \rightarrow 0.617$). The 60k native-objective checkpoint was not evaluated downstream, so that point is missing rather than interpolated, and this is a trend over the checkpoints we measured, not a scaling law.

Alternative explanations were excluded by re-evaluating under matched software, and the training loss is flat, so this is not pixel-space overfitting (Appendix E).

Figure 2b shows the curve under our objective. The panels differ in clip length as well as objective (16 against 32 frames), so the controlled objective comparison is Table 4; what the figure establishes is the direction of each curve.

4.7 SEEDS AND INFERENCE COST

Two seeds of the 60k model were trained and evaluated on the complete six-dataset grid: they give mean MSE/MAE of $0.2852/0.3279$ and $0.2838/0.3272$, a spread of 0.0014 and 0.0007. Table 1 reports the first, which is the weaker of the two. Inference is not single-pass everywhere: the rule averages frame scales on short-horizon cells and rolls out beyond one prediction block, so 9 of the 24 cells cost two or three forward passes (Appendix D), and variants that did not improve on the reported system, including a two-seed prediction ensemble, are listed in Appendix G.

5 DISCUSSION

What the evidence supports, and what bounds it. Two claims stand on separate experiments. A video-pretrained backbone is a better starting point than an image-pretrained one when the adaptation budget is small: at 5k updates, everything but the weights held fixed, it leads by 3–5% at every horizon; and pretrained weights matter far more than either choice, since removing them costs 14% under an otherwise identical trainer. And a transfer framework is what makes any of these initialisations usable: under its own objective the same backbone gets worse at forecasting as that objective improves, and the value-space read-out is worth 5–8% over pixel-only supervision under a

matched ablation. Together they give the system result of Section 4.2; no experiment separates their interaction, so we attribute the end-to-end accuracy to the framework as a whole.

The video advantage closes by 20k updates (Table 3). We keep that beside the claim it bounds, because it says what the video prior supplies: not a higher attainable accuracy but a lower cost of reaching it — the quantity a small adaptation budget is short of, and what makes the headline result a statement about efficiency. At a large adaptation budget, the expectation it sets is that the visual prior matters less than the transfer design.

Limitations that bound the central claim. Our lead is on MSE only: against the same-size VisionTS++ base we take 17 of 24 cells on MSE but 11 on MAE, and the mean MAE is 0.328 against 0.326. The video initialisation’s advantage is a 5k-update result, is gone by 20k, and was never measured at the 60k of the reported system: we did not train a 60k image-initialised counterpart, so nothing here shows that the headline number requires a video initialisation rather than any strong visual prior. The inference rule’s thresholds were also chosen on development runs over four of the six evaluation datasets, so the zero-shot protocol is clean of parameter updates but not of all benchmark knowledge. Two seeds of the main model differ by 0.0014 in mean MSE, which is smaller than our margin over the nearest baseline but of the same order as the differences among the leading models in Table 1. The budget comparison is in corpus size, optimizer updates and sample presentations, not FLOPs, and excludes the upstream video pretraining we inherit. Appendix F collects the caveats that scope the work rather than the claim.

6 CONCLUSION

A video-pretrained backbone can be turned into a competitive zero-shot forecaster on a small time-series budget, but not by substitution: its own objective drives forecasting accuracy the wrong way, and what makes the weights usable is a transfer framework built around the task — period-aligned frames, a predicted block of future frames, a differentiable read-out that places the loss on values, and a training scale distribution matched to the inference rule. One checkpoint under one rule then reaches the lowest mean MSE of the models we compare, after 18.9 GPU-hours on 0.3% of the corpus behind the strongest visual baseline, while staying behind that baseline on mean MAE. The controlled study puts the video prior’s contribution in adaptation efficiency: worth 3–5% at 5k updates and matched by an image prior by 20k. At small adaptation budgets the temporal prior pays; the transfer design is what makes any prior pay at all.

REPRODUCIBILITY STATEMENT

Every number in this paper comes from one publicly documented pipeline. The renderer, the value-space read-out, the trainer and all command-line flags used for each arm are described in Section 3 and Appendix A; the evaluation follows the protocol of Chen et al. (2025) and was checked against their released checkpoint to three decimals (Section 4.1). One checkpoint and one inference rule produce all 24 evaluation cells: no per-dataset tuning, no checkpoint ensembling, no blending with statistical baselines. The pretraining corpus is built from public LOTSA subsets (Woo et al., 2024) with a documented subset list, and none of the six evaluation datasets appears in it. Code, corpus manifest and checkpoints will be released.

ETHICS STATEMENT

The work uses public forecasting benchmarks and a public pretraining corpus; no human subjects, no personally identifying data. Electricity and traffic-like series describe aggregate consumption at substation level. The main risk we see is the usual one for forecasting models sold as general purpose: the reported zero-shot accuracy is measured on six datasets in two domains and should not be read as a guarantee on safety-critical deployments.

USE OF LARGE LANGUAGE MODELS

Large language models were used for copy-editing and for scripting routine experiment orchestration. All experimental design, analysis and claims are the authors’; every number reported here was produced by the pipeline described above.

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A IMPLEMENTATION DETAILS

Rendering. Frames are 224×224 grayscale, replicated to three channels. Fill level 0.12, background 0.92, grid lines every 28 rows at 0.80. Values are standardised with the statistics of the visible context, clipped to $z \in [-3, 3]$ and mapped to height by Equation 1. A clip holds FkP observations in F frames; tubelet size is 2, giving $16 \times 14 \times 14 = 3136$ tokens at $F = 32$. Equation 2 is applied to the mean of the three replicated colour channels. At evaluation the decoded z values are mapped back to the dataset’s scale by $\hat{x} = \hat{z}_{\text{ctx}} + \mu_{\text{ctx}}$, using the same visible-context statistics that produced the rendering.

Training. VideoMAE-B initialised from Kinetics-400, `norm_pix_loss` off, AdamW ($\beta_1=0.9$, $\beta_2=0.95$), learning rate 10^{-4} with cosine decay and 500 warm-up steps, weight decay 0.05 (or 0 with prior anchoring at $\lambda_{\text{sp}} = 0.05$), bf16, micro-batch 8 with 4-step gradient accumulation for an effective batch of 32. Seventy per cent of batches are forecast batches with masked future length h_p drawn from $\{2, 4\}$ frames (1–2 tubelets) and, with probability 0.3, a uniformly drawn $1 \dots (F - h_p - 8)/2$ additional leading tubelets hidden and excluded from the loss; the remaining 30% use tube masking at ratio 0.75 with the pixel term on all masked tokens and no value term. Grid lines are drawn on the background only and are occluded by the fill, so a filled pixel is never a grid pixel. Frame scale $k \in \{1, 2, 4\}$ with probability 0.5/0.3/0.2. Learning rate 2×10^{-4} is worse and 5×10^{-5} ties 10^{-4} .

Corpus. 98 LOTSA subsets, 184,013 series, 654.2M observations, with 20% synthetic series (sinusoids with trend, noise and regime changes) mixed in. Subsets are sampled with probability proportional to the square root of their observation count. We verified that none of ETTh1, ETTh2, ETTm1, ETTm2, electricity or weather is present; the nearest same-domain entries are Monash weather (daily Australian stations, not the Jena 10-minute set) and `traffic_hourly`, which is why traffic is excluded from the evaluation.

Evaluation. `--mode auto` as described in Section 3.5. All 24 cells use the same checkpoint. Multi-channel datasets are forecast channel by channel. Reported MSE and MAE are computed on standardised values over stride-1 test origins.

B PER-CELL COMPARISON

Two baselines publish per-horizon results on this protocol: VisionTS (Table 9 of arXiv 2408.17253v4) and VisionTS++ (Table 4, Appendix C.2 of arXiv 2508.04379v3). Table 6 compares all three on every cell.

Against VisionTS we win 18 of 24 cells on MSE and 18 of 24 on MAE. The six MSE losses are ETTh1 at $H=192/336/720$, ETTh2 at $H=192/336$, and ETTm1 at $H=720$: five are mid-to-long horizons on the hourly sets, the sixth is the longest horizon of a 15-minute set.

Against VisionTS++ base, the same size class as our model, we win 17 of 24 on MSE but only 11 of 24 on MAE, with the MAE losses concentrated on weather, where VisionTS++ wins all four horizons, and on ETTm2. This is the per-cell form of the aggregate MAE gap noted in Section 4.2, and it is the clearest place where our model is behind.

Table 5: Our model, per horizon; same checkpoint and inference rule as Table 1. Six-dataset means in the last row.

Dataset	$H=96$		$H=192$		$H=336$		$H=720$		Avg	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	0.292	0.332	0.332	0.357	0.363	0.377	0.423	0.409	0.352	0.369
ETTm2	0.160	0.245	0.214	0.283	0.264	0.318	0.328	0.362	0.241	0.302
ETTh1	0.352	0.380	0.396	0.408	0.424	0.425	0.446	0.449	0.405	0.416
ETTh2	0.266	0.324	0.334	0.369	0.361	0.395	0.379	0.415	0.335	0.376
ECL	0.128	0.219	0.148	0.239	0.170	0.262	0.207	0.294	0.163	0.254
Weather	0.139	0.183	0.181	0.226	0.232	0.266	0.308	0.331	0.215	0.252
Mean	0.223	0.281	0.267	0.314	0.302	0.341	0.349	0.377	0.285	0.328

Table 6: Per-cell comparison against the two baselines that publish per-horizon results on this protocol: VisionTS (arXiv 2408.17253v4, Table 9) and VisionTS++ base (arXiv 2508.04379v3, Table 4). **Bold**: best of the three in that column group. Counts are given in the text.

Dataset	H	MSE			MAE		
		Ours	VisionTS	VisionTS++ _B	Ours	VisionTS	VisionTS++ _B
ETTm1	96	0.291	0.341	0.316	0.332	0.347	0.343
	192	0.332	0.360	0.347	0.357	0.360	0.362
	336	0.362	0.377	0.368	0.377	0.374	0.379
	720	0.423	0.416	0.408	0.409	0.405	0.405
ETTm2	96	0.160	0.228	0.169	0.245	0.282	0.248
	192	0.214	0.262	0.216	0.283	0.305	0.279
	336	0.264	0.293	0.260	0.318	0.328	0.308
	720	0.328	0.343	0.330	0.362	0.370	0.358
ETTh1	96	0.352	0.353	0.369	0.380	0.383	0.392
	192	0.396	0.392	0.399	0.408	0.410	0.412
	336	0.424	0.407	0.415	0.425	0.423	0.421
	720	0.446	0.406	0.424	0.449	0.441	0.437
ETTh2	96	0.266	0.271	0.277	0.324	0.328	0.326
	192	0.334	0.328	0.333	0.369	0.367	0.362
	336	0.361	0.345	0.350	0.395	0.381	0.384
	720	0.379	0.388	0.370	0.415	0.422	0.409
ECL	96	0.128	0.177	0.152	0.219	0.266	0.237
	192	0.148	0.188	0.168	0.239	0.277	0.252
	336	0.170	0.207	0.186	0.262	0.296	0.269
	720	0.207	0.256	0.228	0.294	0.337	0.303
Weather	96	0.139	0.220	0.145	0.183	0.257	0.179
	192	0.181	0.244	0.187	0.226	0.275	0.219
	336	0.232	0.280	0.240	0.266	0.299	0.258
	720	0.308	0.330	0.317	0.331	0.337	0.308

C INITIALISATION AT TWO CLIP LENGTHS

At 16 frames the video prior is worth 3–4%; at 32 frames the two initialisations tie. Read together with the anchoring ablation (inert at 32 frames, essential at 16), this suggests that part of what video pretraining supplies is long-range temporal context, which a longer window supplies directly. The matched-size comparison in Table 3 therefore uses the 5k-step budget, where the prior is still doing work.

Table 7: Video against image initialisation at two clip lengths, two seeds each where available. Five-dataset mean MSE at $H=96/336/720$.

Clip length	Video init	Image init
16 frames	0.257/0.350/0.383, 0.256/0.348/0.384	0.268/0.359/0.400, 0.265/0.353/0.400
32 frames	0.245/0.330/0.379	0.244/0.329/0.373, 0.245/0.330/0.373

D INFERENCE CONFIGURATION AND COST

The rule of Section 3.5 maps (P, H) to a configuration with no per-dataset tuning. Table 8 lists the configuration it selects for each evaluation cell and the number of forward passes that configuration costs. `mp` (multi-period) predicts one block of frames with k periods per frame; the `sc` suffix lists the frame scales that are averaged, each of which is a separate pass; `rollout2` feeds the prediction back and predicts twice. The three $P=24$ datasets share a configuration, as the rule depends only on (P, H, L) . Nine of the 24 cells use more than one pass; the other fifteen use one.

Table 8: Inference configuration and forward passes per evaluation cell.

Dataset	P	$H=96$	$H=192$	$H=336$	$H=720$
ETTh1 / ETTh2 / ECL	24	sc124 (3)	sc12 (2)	mp (1)	rollout2 (2)
ETTm1 / ETTm2	96	mp (1)	mp (1)	mp (1)	mp (1)
Weather	144	mp (1)	mp (1)	mp (1)	mp (1)

A 32-frame clip costs roughly $8 \times$ a 16-frame clip per pass on the largest dataset, so the inference cost of the reported system is higher than that of a 16-frame configuration even where the pass count is one.

E CHECKS ON THE OBJECTIVE-MISMATCH RESULT

The degradation in Figure 2a survives the following checks. Checkpoint key sets, the `transformers` version, the attention configuration and dataloader warnings were excluded by re-evaluating every checkpoint under matched software. The training loss is flat across the range $(0.0092 \rightarrow 0.0089)$, so the model is not overfitting the pixel objective. Per cell, the 176k-step model is 5–60% worse than the 20k-step model, with the largest gaps at the longest horizons (ETTh2 at $H=720$: $0.429 \rightarrow 0.617$). The one quantity that moves monotonically with the schedule is the weight norm, which shrinks by roughly 30% over the $9 \times$ longer run under decoupled weight decay; this is consistent with the prior-anchoring ablation helping in the short-clip regime, though we did not isolate it as the cause. The 60k native-objective checkpoint was never evaluated downstream, so no value is reported for it.

F SCOPE AND CAVEATS

Where the model loses. Mid-to-long horizons on the hourly ETT sets stay behind the strongest baselines, because the context is capped at 32 frames of at most four periods: covering $H=720$ with $P=24$ trades phase resolution for coverage.

Seeds. Two seeds of the 60k model were trained and evaluated on all 24 cells. Their six-dataset four-horizon means are 0.2852/0.3279 (seed 0) and 0.2838/0.3272 (seed 1). Per dataset the two agree to within 0.005 on MSE everywhere except ETTm1 (0.3523 against 0.3472). Table 1 reports seed 0, the weaker of the two; the two-seed mean would be 0.2845/0.3276. We report a single seed’s full grid rather than an average so that every number in the paper comes from one checkpoint under one inference rule.

How the inference rule was fixed. The rule contains no dataset identity and all 24 main-table cells were produced by it. Its two thresholds — keep $kP \leq 96$ samples per frame, and roll out only when $P < 96$ — were chosen by comparing configurations on development runs over four of the six datasets, so the rule is dataset-agnostic in form but was not selected blind to the benchmark.

Corpus composition and overlap. The corpus was built without domain exclusion. It contains no ETT data of any kind. It does contain Monash weather (2,850 series, 40.6M observations; daily Australian stations, a different source from the 10-minute Jena set used as the LSF weather benchmark) and Monash traffic_hourly (820 series, 14.2M observations; the same freeway-occupancy source as the LSF traffic benchmark), along with PEMS_*, LOOP_SEATTLE, lcl and london_smart_meters. Traffic is therefore excluded from every table in this paper. An exact-match anchor screen of the benchmark files against the corpus returned no matches, but it was run against an earlier, domain-excluded build (278.2M observations) rather than the 654.2M build these models used; repeating it on the build in use is the first item in our list of open checks. The screen also cannot detect rescaled, aggregated or interpolated copies.

Architecture matching. The ViT-B initialisation comparison is size-matched rather than architecturally exact: the V-JEPA arms carry an added predictor, an EMA target encoder, a latent loss term and a re-initialised read-out head, and they do not train on tube-masked batches.

G ALTERNATIVES THAT DID NOT WORK

We list these so that the obvious questions have answers, not because any of them is a finding.

Table 9: Attempts that did not improve on the recipe in Section 3.

Attempt	Outcome
9× budget under the pixel objective (176k steps)	5–60% worse per cell
Larger batch (128, 256 via accumulation)	equal or worse at every horizon
4× more in-domain data (buildings_900k, LargeST, Q-Traffic, Azure)	ties the corpus we use
Longer masked future (8 of 16 frames)	slightly worse
Value-loss weight 0.3 instead of 0.1	worse
Three-checkpoint ensemble	no gain
Blending with a zero-parameter statistical prior	validation selection does not transfer
Weight-space averaging of two seeds	better on ETT, 16% worse on electricity, net worse
Cross-corpus weight averaging	worse than either parent

Prediction-space averaging of two seeds is the one combination that helps, by 0.6% and never worse on any cell; it appears as a secondary row in Table 1.